

Efficacy and safety of ablation index-guided catheter ablation for atrial fibrillation: an updated meta-analysis

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Received 1 April 2020; editorial decision 5 July 2020

Aims

Despite recent advances in catheter ablation for atrial fibrillation (AF), pulmonary vein reconnection (PVR), and AF recurrence remain significantly high. Ablation index (AI) is a new method incorporating contact force, time, and power that should optimize procedural outcomes. We aimed to evaluate the efficacy and safety of AI-guided catheter ablation compared to a non-AI-guided approach.

Methods and results

A systematic search was performed on MEDLINE (via PubMed), EMBASE, COCHRANE, and European Society of Cardiology (ESC) databases (from inception to 1 July 2019). We included only studies that compared AI-guided with non-AI-guided catheter ablation of AF. Eleven studies reporting on 2306 patients were identified. Median follow-up period was 12 months. Ablation index-guided ablation had a significant shorter procedural time (141.0 vs. 152.8 min, $P = 0.01$; $I^2 = 90\%$), ablation time (21.8 vs. 32.0 min, $P < 0.00001$; $I^2 = 0\%$), achieved first-pass isolation more frequently [odds ratio (OR) = 0.09, 95%CI 0.04–0.21; 93.4% vs. 62.9%, $P < 0.001$; $I^2 = 58\%$] and was less frequently associated with acute PVR (OR = 0.37, 95%CI 0.18–0.75; 18.0% vs. 35.0%; $P = 0.006$; $I^2 = 0\%$). Importantly, atrial arrhythmia relapse post-blanking was significantly lower in AI compared to non-AI catheter ablation (OR = 0.41, 95%CI 0.25–0.66; 11.8% vs. 24.9%, $P = 0.0003$; $I^2 = 35\%$). Finally, there was no difference in complication rate between AI and non-AI ablation, with the number of cardiac tamponade events in the AI group less being numerically lower (OR = 0.69, 95%CI 0.30–1.60, 1.6% vs. 2.5%, $P = 0.39$; $I^2 = 0\%$).

Conclusions

These data suggest that AI-guided catheter ablation is associated with increased efficacy of AF ablation, while preserving a comparable safety profile to non-AI catheter ablation.

Keywords

Ablation index • Atrial fibrillation • Radiofrequency ablation • Meta-analysis

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What's new?

- Ablation index (AI) is a commercially available new marker/index that combines contact force, radiofrequency time, and radiofrequency power in a non-linear formula in real time.
- It is suggested to allow intra-procedural visualization of radiofrequency current application in a 3D mapping system, therefore improving outcomes in patients undergoing atrial fibrillation ablation.
- In this meta-analysis, we demonstrated that AI-guided catheter ablation had a shorter total procedural time, achieved first-pass isolation more frequently and atrial arrhythmia relapse post-blanking was significantly lower, with no difference in complication rates.
- A large randomized-controlled trial is required to confirm the benefits of AI-guided catheter ablation.

Introduction

Pulmonary vein isolation (PVI) is well-established as the gold standard for ablation of paroxysmal atrial fibrillation (AF) and has been shown to be non-inferior to extensive ablation in persistent AF.^{1–3}

Despite recent advances in technology and ablation methods such as the use of contact force (CF) and force-time integral (FTI), AF recurrence rates remain significantly high for both paroxysmal and persistent AF.^{4,5} Studies have shown that low CF and FTI during AF ablation are associated with a higher incidence of PV reconnection (PVR) and recurrence of AF.^{6,7} Contact force sensing catheters are now routinely used, but FTI has failed to make a clear impact on AF ablation outcomes⁸ and hence alternatives are being developed. Radiofrequency (RF) power is still not taken into account by FTI.⁹ Furthermore, FTI assumes a linear association between CF and time, which is not the case according to recently published data.¹⁰

Ablation index is a commercially available new marker/index that combines CF, RF time, and RF power in a non-linear formula in real time.^{11–13} It is suggested to allow intra-procedural visualization of radiofrequency current application in a 3D mapping system, therefore improving outcomes in patients undergoing AF ablation.

In this meta-analysis, we aim to assess the efficacy and safety profile of AI in patients undergoing catheter ablation for AF compared those having a non-AI procedure.

Methods

Study selection

A systematic electronic search was performed on PubMed, EMBASE, Cochrane clinical trials database (from inception to July 2019) with no language limitations, using the following search string: 'atrial fibrillation' AND ('ablation' OR 'catheter ablation' OR 'radiofrequency ablation') AND 'ablation index'.

The population, intervention, comparison, and outcome (PICO) approach was used. The population of interest included patients with AF, and the intervention was catheter ablation of AF comprising of PVI ($\pm$ additional lesions) using radiofrequency. The comparison of interest was AI-guided ablation vs. non-AI guided ablation. The primary outcome measure was AF or atrial tachycardia recurrence post-blanking. Other

outcomes included PV isolation during first-pass, acute PV reconnection, total procedural time, ablation time, and fluoroscopy time. Assessed procedural complications were cardiac tamponade, a composite of all pericardial complications, oesophageal fistula, phrenic nerve palsy, stroke, and 'other life-threatening complications', assessed on a study-by-study basis.

In order to be included, studies needed to provide a minimum of information regarding the patient demographics of the sample of AF patients undergoing catheter ablation, and at least one of the aforementioned outcomes of interest. Observational non-controlled case series required a minimum of five patients to be considered eligible.

For the endpoint Relapse after blanking, the minimum mean study follow-up duration was 12 months.

Review articles, editorials, and case reports were not considered eligible for the purpose of this review.

Three independent reviewers (A.I., T.W., and R.P.) screened all abstracts and titles to identify potentially eligible studies. The full texts of these potentially eligible studies were then evaluated. A.I. and W.L. formally evaluated study quality using the Ottawa–Newcastle scale.¹³ Agreement between both reviewers was required for decisions regarding inclusion or exclusion of studies. When this was not possible, RP intervened as the arbiter and had the final decision. Data extraction and presentation for the preparation of this manuscript followed the recommendations of the PRISMA group (Figure 1). The following data were extracted for characterizing each patient sample in the selected studies, whenever available: demographics and sample characterization, AF duration, presence of structural heart disease, atrial size, left ventricular ejection fraction and proportion of persistent and paroxysmal AF patients (Table 1 and Supplementary material online, Table S1), ablation settings (Table 2), procedural information (Table 3) blanking period, follow-up duration, monitoring of AF relapse, and use of anti-arrhythmic drugs (Table 4).

The definitions AF/atrial arrhythmia relapse, blanking period, and methods used for monitoring during follow-up were collected in all studies.

Reference lists of all accessed full-text articles were further searched for sources of potentially relevant information. Authors of full-text papers were also contacted by email to retrieve additional information.

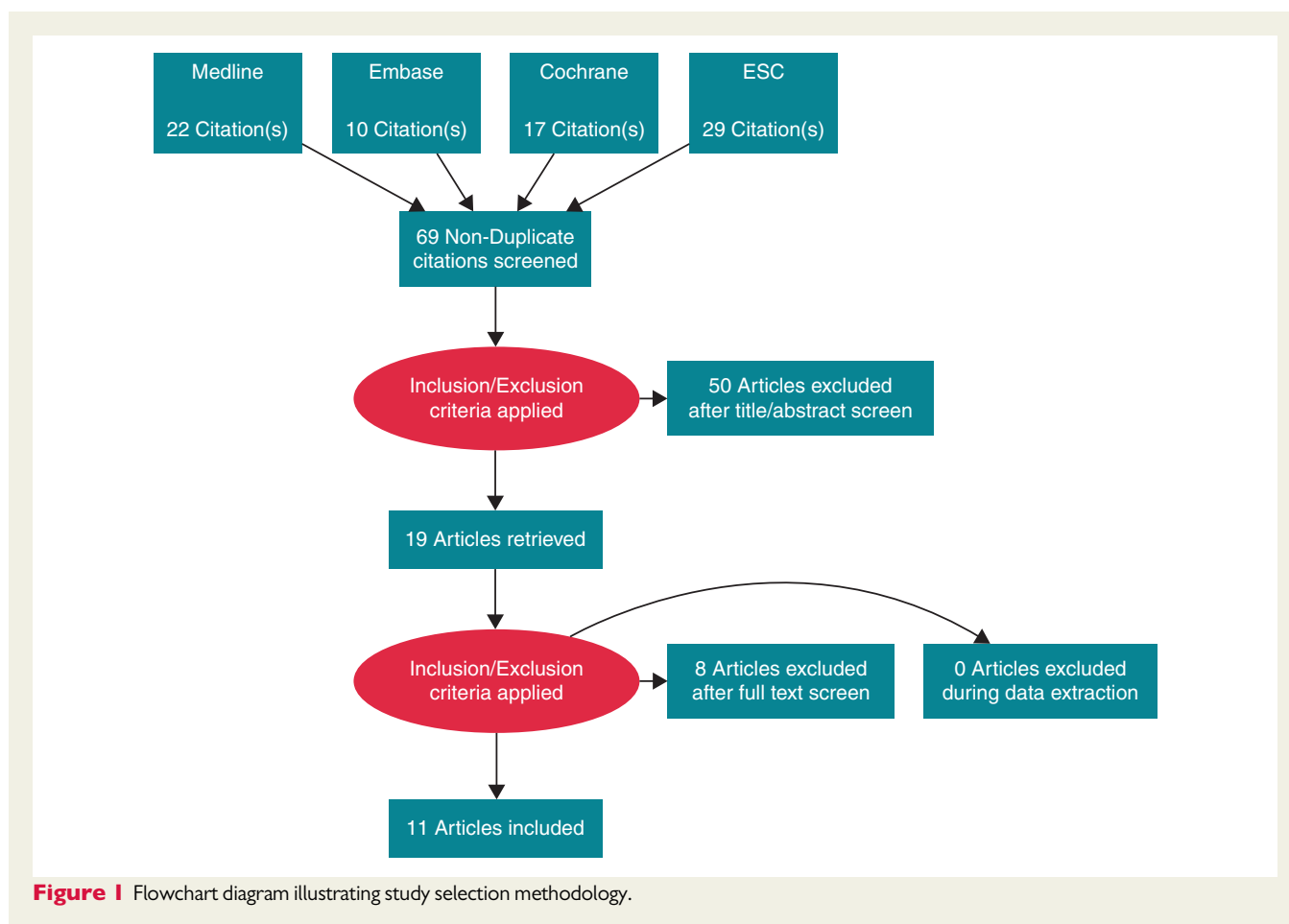
Statistical analysis

Data were pooled using random effects, according to the Mantel-Haenszel model, through Review Manager (RevMan), Version 5.1. (Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration, 2011). The odds ratio (OR) and respective 95% confidence intervals (95%CI) were used as a measurement of treatment effect for binary outcomes, and the mean difference and respective 95%CI were used as a measure of treatment effect for continuous outcomes. Pairwise comparisons were performed all the endpoints.

The number needed to treat (NNT) was estimated for the primary endpoint: number of patients who have to be treated with the active treatment group to prevent a relapse.

In order to assess study design-related factors that could interfere with the results of the meta-analysis, sub-analyses were performed for the primary efficacy endpoint to assess the impact of study design, proportion of patients with paroxysmal AF, type of ablation catheters used, and date when studies were published.

Statistical heterogeneity on each outcome of interest was assessed and quantified using the Cochran Q test and the I^2 statistic, respectively. The I^2 statistic describes the percentage of total variation across studies due to heterogeneity rather than chance. Values of <5%, 25–50%, and



>50% are by convention classified low, moderate, and high degrees of heterogeneity, respectively.

Results

Search results

A total of 69 non-duplicate citations were screened of which 50 were excluded following screening of the title and abstract. The remaining 19 results were carefully screened, and after analysis of the full-text, only 11 articles^{14–24} were considered appropriate for the purpose of our meta-analysis. The selection process is illustrated in Figure 1 and there was an excellent agreement between investigators on the inclusion of the selected trials.

Study design and population

The baseline data of patients and the design of the trials are summarized in Table 1 and Supplementary material online, Table S1. A total of 2306 patients were included, 1046 (45.4%) underwent AI-guided ablation and 1260 (54.5%) underwent non-AI (control group). Four of the studies were retrospective and seven were prospective studies. None of the studies was randomized controlled. The type of AF was recorded for 2096 patients, 1215 (58.0%) of which had paroxysmal AF.

Quality assessment of the included studies was assessed using the Newcastle–Ottawa score, as they were all cohort studies and demonstrated that three studies scored 9,^{14,18,20} four studies scored 8,^{16,19,21,23,24} two studies scored 6,^{17,22} and one study scored 5¹⁵ (Supplementary material online, Table S2). The Delphi criteria were not used, as none of the studies was randomized-controlled trials.

Funnel plots were carried out on the comparison for procedural time only, as there were not 10 studies available for any of the other comparisons.

Ablation index and ablation settings

Most studies used lower power levels for the posterior wall (Table 2). However, Hussein *et al.*, Philips *et al.*, and Kobori *et al.* used the same power levels all around the left atrium (30, 35, and 50 W, respectively). Dhillon *et al.* utilized 40 W for the anterior wall. Settings for catheter stability varied widely going from 2.5 to 4 mm and from 3 to 15 s. Minimum force was 5 g for most studies, despite presenting variation. Similarly, force over-time varied widely, between 25% and 60%, with 25% and 30% being the most popular settings. Maximum interlesion distance was set as ≤6 mm for most studies (six out of eight studies). Ablation index targets were lower for the posterior wall, usually set between 380 and 450, and for the anterior wall, it ranged between 400 and 600 (Table 2).

Table 1 Study information and baseline characteristics

References	Design	N	Indication	AF type	AF duration (months)	AADs	Age (years)	Males	CHA ₂ DS ₂ VASc	Structural heart disease	Left atrial diameter (mm)	LVEF
Hussein et al. ¹⁴	Prospective analysis Prop-matched controls Single centre	AI—89 Ctrl—9	Symptomatic AF or refractory to drug therapy	AI PAF—43 (48%) AI PeAF—31 (35%) AI LSPeAF—15 (17%) Ctrl PAF—45 (50%) Ctrl PeAF—29 (33%) Ctrl LSPeAF—15 (17%)	AI PeAF—7 (5–10) AI LSPeAF—22 (12–28) Ctrl PeAF—9 (5–12) Ctrl LSPeAF—24 (18–33)	N/A	AI—62 ± 9.1 Ctrl—62 ± 9.1	AI—67 (75%) Ctrl—65 (73%)	CHADs ₂ Score AI—0.65 ± 0.77 Ctrl—0.65 ± 0.93	N/A	AI—42 ± 6 Ctrl—41 ± 6	N/A
Pontoppidan et al. ^{15a}	Prospective Controls were consecutive cases Single centre	AI—67 Controls were consec- Ctrl—64	Symptomatic AF	AI PAF—49 (73%) Ctrl PAF—23 (64%)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Münkler et al. ¹⁶	Retrospective analysis Controls were consecutive cases Single centre	AI—23 Controls were consec- Ctrl—11	Symptomatic AF or refractory to drug therapy	AI PAF—23 (100%) Ctrl PAF—11 (100%)	N/A	AI and Ctrl Amio—9 (26.5%) BB—29 (85.3%) Ic—15 (44.1%) CCB—11 (32.3%)	AI and Ctrl—63.7 ± 11.5 (65%)	AI and Ctrl—22 (65%)	AI and Ctrl—<2—13 (38.2%) 2–4—6 (17.6%) >4—4 (11.8%)	N/A	AI and Ctrl - 37.5 ± 3.8	AI and Ctrl—54.5 ± 3.5
Neven et al. ^{17a}	Prospective study Controls were consecutive cases Single centre	AI—387 Controls were consec- Ctrl—686	Drug refractory AF	AI and Ctrl PAF—444 (41%)	N/A	N/A	AI and Ctrl—67 ± 11 (59%)	AI and Ctrl—633 (59%)	N/A	N/A	N/A	N/A
Philips et al. ¹⁸	Prospective study Controls were the last consecutive non-AI cases Single centre	AI—50 Controls were the last Ctrl—50	Symptomatic PAF	AI PAF—50 (100%) Ctrl PAF—50 (100%)	AI—24 (12–84) Ctrl—24 (12–72)	N/A	AI—61 ± 11 Ctrl—62 ± 10	AI—36 (72%) Ctrl—31 (62%)	CHA ₂ DS ₂ VASc score ≥ 2 AI—16 (32%) Ctrl—25 (50%)	AI—3 (6%) Ctrl—7 (14%)	AI—39.9 ± 3.9 Ctrl—40.4 ± 4.4	AI—All normal except for one that was 50% Ctrl—All normal
Solimene 2018 ¹⁹	Prospective study Controls were consecutive cases from a previous study Multi-Centre	AI—156 Controls were consec- Ctrl—120	Symptomatic AF or refractory to drug therapy	AI PAF—124 (79.5%) AI PeAF—32 (20.5%) Ctrl—N/A	N/A	AI Amio—17 (11%) AI Ic—7 (4%) Ctrl—N/A	AI—50 ± 10 Ctrl—56 ± 11	AI—76 (49%) Ctrl—83 (69%)	N/A	AI—68 (44%) Ctrl—54 (45%)	AI—45 ± 9 Ctrl—46 ± 11	AI—57 ± 6 Ctrl—57 ± 7
Casella et al. ²⁰	Prospective study Propensity matched controls Single centre	AI—72 Ctrl—72	N/A	AI PAF—48 (67%) AI PeAF—24 (33%) Ctrl PAF—46 (64%) AI PeAF—26 (36%)	N/A	AI Amio—19 (26%) AI AAD—43 (60%) Ctrl Amio—42.5 ± 7.8 Ctrl AAD—40.8 ± 15.1	AI—59 ± 14 Ctrl—61 ± 9	AI—49 (68%) Ctrl—57 (79%)	N/A	N/A	AI—42.4 ± 10.9 Ctrl—42.5 ± 7.8	N/A

Continued

Table 1 Continued

References	Design	N	Indication	AF type	AF duration (months)	AADs	Age (years)	Males	CHA ₂ DS ₂ -VASc	Structural heart disease	Left atrial diameter (mm)	LVEF
Dhillon <i>et al.</i> ²¹	Prospective study Controls were consecutive cases in the same centres before the introduction of AI	AI—50 Ctrl—50	Symptomatic AF or refractory to drug therapy	AI PAF—50 (100%) Ctrl PAF—50 (100%)	AI—24 (17–48) Ctrl—42 (19–71)	N/A	AI—60.1 ± 11.8 Ctrl—59.9 ± 10.8	AI—35 (70%) Ctrl—24 (48%)	AI—1.3 ± 1.2 Ctrl—1.68 ± 1.6	AI—8 (16%) Ctrl—N/A	AI—37.6 ± 0.5 Ctrl—38.7 ± 0.4	AI—56.0 ± 7.3 Ctrl—N/A
Kobori <i>et al.</i> ^{22a}	Multi-centre Prospective study Controls were consecutive cases Single centre	AI—80 (40 high power and 40 low power) Ctrl—40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Lee <i>et al.</i> ²³	Retrospective study Controls were consecutive cases Single centre	AI—32 Ctrl—38	Drug refractory AF	AI PAF—24 (78.1%) Ctrl PAF—23 (60.5%)	N/A	N/A	AI—59.9 ± 9.1 Ctrl—58.6 ± 7.9	AI—25 (78.1%) Ctrl—30 (78.9%)	AI—1.3 ± 1.4 Ctrl—1.2 ± 1.2	N/A	AI—41.2 ± 5.4 Ctrl—41.8 ± 8.7	N/A
Parollo <i>et al.</i> ^{24a}	Retrospective Matched but criteria for matching not specified Single Centre	AI—40 Ctrl—40	N/A	AI PAF—40 (100%) Ctrl PAF—40 (100%)	N/A	N/A	AI—62.4 ± 9.5 Ctrl—61.8 ± 7.0	AI—22 (56%) Ctrl—18 (44%)	N/A	N/A	AI—40.2 ± 5.9 Ctrl—39.2 ± 6.5	N/A

^aIndicates studies that were published only as abstracts.

AAD, anti-arrhythmic drug; AF, atrial fibrillation; AI, ablation index guided catheter ablation group; Amio, amiodarone; BB, beta-blocker; CCB, calcium channel blocker; Ctrl, control group (non-AI guided catheter ablation); Ic, class Ic anti-arrhythmic; LSPeAF, long standing persistent atrial fibrillation; LVEF, left ventricular ejection fraction; PAF, paroxysmal atrial fibrillation; PeAF, persistent atrial fibrillation.

Table 2 Ablation settings

References	Electroanatomical mapping tools	Ablation catheters	Catheter stability	Minimum CF	Force over-time	Maximum interlesion distance	Energy (W)	AI Target	Minimum AI value in PVR group	Minimum AI value in no-PVR group
Hussein et al. ¹⁴	CARTO 3 (Biosense Webster ©) integration with CT or MRI reconstruction of the left atrium (CartoMerge, Biosense Webster ©)	Thermocool SmartTouch (Biosense Webster ©)	3 mm for 8 s	5 g	30%	≤5 mm	Ant—35 Roof—35 Post—35 Inf—35	Ant—537 Roof—537 Post—420 Inf—420	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A
Pontoppidan et al. ¹⁵	CARTO 3 v6 (Biosense Webster ©)	N/A	3 mm for 6 sec	3 g	25%	≤6 mm	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—500 Roof—500 Post—400 Inf—400	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A
Münkler et al. ¹⁶	CARTO 3 (Biosense Webster ©)	ThermoCool SmartTouch SF (Biosense Webster ©)	3 mm for 3 s	3 g	25%	N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—500 Roof—450 Post—380 Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A
Neven et al. ¹⁷	CARTO 3	ThermoCool SmartTouch SF (Biosense Webster ©) for AI group Non-contact force-sensing catheter for control group	N/A	N/A	N/A	N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—600 Roof—450 Post—450 Inf—450	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A
Philips et al. ¹⁸	CARTO 3 (Biosense Webster ©)	ThermoCool SmartTouch (Biosense Webster ©)	3 mm for 8 s	4 g	30%	≤6 mm	Ant—30 Roof—30 Post—30 Inf—30	Ant—550 Roof—400 Post—400 Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A
Solimene 2018	CARTO 3 v4 (Biosense Webster ©)	ThermoCool SmartTouch SF (Biosense Webster ©)	3 mm for 3 s	5 g	30%	≤6 mm	Ant—35 Roof—35 Post—25 Inf—25	Ant—400–450 Roof—330–350 Post—330–350 Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A
Casella et al. ²⁰	CARTO 3 (Biosense Webster ©)	ThermoCool SmartTouch SF (Biosense Webster ©)	3 mm for 3 s	3 g	25%	≤6 mm	Ant—35 Roof—35 Post—30 Inf—30	Ant—500–550 Roof—N/A Post—400 Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A
Dhillon et al. ²¹	CARTO 3 (Biosense Webster ©)	ThermoCool SmartTouch SF (Biosense Webster ©) in the high power ablation index (HPAI) group	4 mm for 5 s	5 g	50%	≤6 mm	Ant—40 Roof—40 Post—30 Inf—40	Ant—450 Roof—450 Post—350 Inf—450	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A

Continued

Table 2 Continued

References	Electroanatomical mapping tools	Ablation catheters	Catheter stability	Minimum CF	Force over-time	Maximum interlesion distance	Energy (W)	AI Target	Minimum AI value in PVR group	Minimum AI value in no-PVR group
ThermoCool SmartTouch (Biosense Webster ©) in the control group										
Kobori et al. ^{22a}	CARTO 3	N/A	N/A	N/A	N/A	N/A	Ant—FTI and LP-AI (35), HPAL (50) Roof—FTI and LP-AI (35), HPAL (50) Post—FTI and LP-AI (35), HPAL (50)	Ant—550 Roof—550 Post—400 Inf—400	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A
Inf—FTI and LP-AI (35), HPAL (50)										
Lee et al. ²³	CARTO 3 (Biosense Webster ©)	CF-sensing catheter (CON group).	2.5 mm for 5 s	5 g	25%	≤4 mm	Ant—25–30 Roof—25–30 Post—25 Inf—25	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—450 Roof—450 Post—350 Inf—350
Parollo et al. ²⁴	CARTO 3	N/A	2.5 mm for 15 sec	5 g	60%	≤6 mm	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—500 Roof—500 Post—380 Inf—380	Ant—N/A Roof—N/A Post—N/A Inf—N/A	Ant—N/A Roof—N/A Post—N/A Inf—N/A

AI, ablation index; Ant, anterior; CF, contact force; CT, computed tomography; Inf, inferior; MRI, magnetic resonance imaging; Post, posterior; PVR, pulmonary vein reconnection.

Table 3 Procedural information

References	Total Procedure Time (min)	Total Fluoroscopy Time (min)	Ablation time (min)	PVI after single encirclement	Acute PVR	Impedance drop	Procedural Complications
Hussein et al. ¹⁴	AI—175 ± 31 Ctrl—163 ± 47	AI—11.9 ± 7.7 Ctrl—11.8 ± 5.6	AI—N/A Ctrl—N/A	AI—173/178 Ctrl—149/177	Number of circles with acute PVR AI—11/183 Ctrl—24/218	AI—13.7 [9–19] Ω Ctrl—8.8 [5.2–13] Ω	AI—0/89 Ctrl—2/89 (1 phrenic nerve palsy and 1 retroperitoneal haematoma)
Pontoppidan et al. ¹⁵	AI—115.5 ± 31.1 Ctrl—115.4 ± 33.9	AI—9.6 ± 4.3 Ctrl—15 ± 7.8	AI—N/A Ctrl—N/A	AI—41/67 Ctrl—N/A	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	AI—0/67 Ctrl—0/64
Münkler et al. ¹⁶	AI—N/A Ctrl—N/A	AI—15.5 ± 6.2 Ctrl—N/A	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	AI—0/23 Ctrl—0/11
Neven et al. ¹⁷	AI—164 ± 51 Ctrl—186 ± 53	AI—5.0 ± 3.0 Ctrl—7.0 ± 5.0	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	AI—7/387 Ctrl—17/686 (all tamponade)
Philips et al. ¹⁸	AI—149 ± 33 Ctrl—192 ± 42	AI—26.3 ± 7.8 Ctrl—21.3 ± 9.0	AI—18 ± 4 Ctrl—28 ± 7.5	AI—98/100 Ctrl—54/100	Number of patients with acute PVR AI—2/50 Ctrl—9/50	AI—N/A Ctrl—N/A	AI—0/50 Ctrl—1/50 (tamponade)
Solimene 2018	AI—95 ± 30 Ctrl—96 ± 36	AI—5.0 ± 6.0 Ctrl—7.0 ± 6.0	AI—26 ± 10 Ctrl—N/A	AI—568/628 Ctrl—N/A	AI—47/627 Ctrl—N/A	AI—impedance drop ≥ 10 Ω 57.1% (5790/10141) of RF pulses Ctrl—N/A	AI—3/156 (1 pericardial effusion and 2 haematomas—none required intervention) Ctrl—N/A
Casella et al. ²⁰	AI—120 ± 34 Ctrl—119 ± 39	AI—15.0 ± 8.5 Ctrl—19.7 ± 10.5	AI—28.2 ± 9.5 Ctrl—39 ± 9.6	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	AI—0 Ctrl—0
Dhillon et al. ²¹	AI—159.06 ± 38.82 Ctrl—206.33 ± 41.88	AI—7.67 ± 4.72 Ctrl—12.78 ± 1.72	AI—28.88 ± 9.43 Ctrl—45.82 ± 15.72	AI—82/100 Ctrl—34/100	Number of veins with acute PVR AI—28/200 (14%) Ctrl—48/200 (28%)	AI—N/A Ctrl—N/A	AI—0/50 Ctrl—3/50 (2 episodes of pericarditis and 1 femoral venous haematoma)
Kobori et al. ²²	AI—high power 68.4 ± 7.0 and low power = 90.6 ± 4	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	AI—N/A Ctrl—N/A	Number of patients with acute PVR AI—16/50 Ctrl—26/50	AI—N/A Ctrl—N/A	AI—0/80 Ctrl—0/40

Continued

Table 3 Continued

References	Total Procedure Time (min)	Total Fluoroscopy Time (min)	Ablation time (min)	PVI after single encirclement	Acute PVR	Impedance drop	Procedural Complications
Lee et al. ²³	Ctrl—96.3±5.8						
	AI—199.1±41.6	AI—19.8±7.3	AI—77.7±20.1	AI—N/A	Number of segments with PVR (each patient had 14 segments)	AI—13.0±4.4 Ω	N/A
	Ctrl—223.4±48.5	Ctrl—24.7±25.9	Ctrl—87.1±17.8	Ctrl—N/A	AI—19/448 (4.2%) Ctrl—57/532 (10.7%)	Ctrl—9.9±3.5 Ω	
Parollo et al. ²⁴	AI—195.88±44.00	AI—9.56±4.69	AI—N/A	AI—N/A	AI—N/A	AI—N/A	AI—2/40
	Ctrl—178.57±33.50	Ctrl—11.16±7.11	Ctrl—N/A	Ctrl—N/A	Ctrl—N/A	Ctrl—N/A	Ctrl—2/40 (1 pericardial effusion not requiring drainage and 1 arteriovenous fistula in each group)

AI, ablation index guided catheter ablation group; Ctrl, control group (non-ablation index guided catheter ablation); PVI, pulmonary vein isolation; PVR, pulmonary vein reconnection.

Acute procedural outcomes

Procedural time was available in 10 studies. Ablation index-guided ablation had a statistically significant shorter procedural time (141.0 min vs. 152.8 min, $P=0.01$) with high heterogeneity across studies ($I^2=90\%$, $P<0.001$). Funnel plots suggested potential publication bias regarding this endpoint with most of the study results outside the specified 95%CI in the plot (Figure 2). In addition, AI-guided ablation had a significantly shorter ablation time (21.8 min vs. 32.0 min, $P<0.00001$; very low heterogeneity, $I^2=0\%$) (Supplementary material online, Figure S1) and fluoroscopy time (10.2 min vs. 12.0 min, $P=0.009$; $I^2=75\%$) compared to non-AI-guided ablation (Supplementary material online, Figure S2).

Procedures performed using AI-guided radiofrequency ablation had a higher rate of first-pass PVI (OR = 0.09, 95%CI 0.04–0.21; 93.4% vs. 62.9%, $P<0.001$; high heterogeneity, $I^2=58\%$) (Figure 3). Importantly, acute PVR was less likely to occur in the AI (OR = 0.37, 95%CI 0.18–0.75; 18.0% vs. 35.0%; $P=0.006$, very low heterogeneity, $I=0\%$) (Supplementary material online, Figure S3).

In all but one of the studies, ablation procedure complications were reported. These included pericardial tamponade, pericardial effusion (not requiring drainage), arteriovenous fistula, phrenic nerve palsy, and retroperitoneal haematoma. Cardiac tamponade as the most common complication was reported in 7 patients undergoing AI-guided catheter ablation compared to 17 in the non-AI studies (1.6% vs. 2.4%, $P=0.39$) (Supplementary material online, Figure S4), while pericardial complications (including both pericardial tamponade and pericardial effusion not requiring drainage) were reported in 8 patients undergoing AI-guided catheter ablation compared to 21 in the non-AI studies (1.5% vs. 2.5%, $P=0.28$) (Supplementary material online, Figure S5). Of note, there were no atrio-oesophageal fistulas, strokes, or deaths reported in either group.

Freedom from atrial fibrillation/at relapse post-blanking

Recurrence of indexed arrhythmia was evaluated by clinical assessment, electrocardiogram or Holter. Follow-up duration was available in six of the studies (blanking period 3 months). The overall median follow-up was 12 months, while the interquartile range was 12.0–16.5 months. The minimum median follow-up was 12 months, while the maximum median follow-up was 24 months.

Analysis of available data from six studies showed that AI-guided catheter ablation is characterized by significantly less AF recurrence compared to non-AI ablation studies (OR = 0.41, 95%CI 0.25–0.66; 11.8% vs. 24.9%; $P=0.0003$; moderate heterogeneity, $I^2=35\%$) (Figure 4). The number of patients who had to be treated (NNT) with the AI-guided strategy to prevent one relapse (compared to control) was 7.6.

Sub-analyses for this primary endpoint based on different scenarios are shown in the Supplementary material online, Table S3. For most of the assessed sub-analyses, except for retrospective studies or studies published only as an abstract,²⁴ the AI-guided approach was associated with a significant reduction in atrial arrhythmia relapse.²⁴ Multi-centre studies and those where the SmartTouch SF catheter was used displayed higher levels of heterogeneity.

Table 4 Procedural outcomes

References	Blanking Period Duration	Follow-up Duration (months)	Method of follow-up	AT/AF Relapse after blanking	AADs post-ablation
Hussein et al. ¹⁴	3 months	12	3 monthly 12-lead ECGs, and symptom determined 24-h Holter ECG monitoring as needed	AI—15 (17%) Ctrl—39 (37%)	None
Pontoppidan et al. ¹⁵	N/A	N/A	N/A	N/A	N/A
Münkler et al. ¹⁶	N/A	N/A	N/A	N/A	N/A
Neven et al. ¹⁷	N/A	N/A	N/A	N/A	N/A
Philips et al. ¹⁸	3 months	12	Clinical evaluation and 12-lead ECG at 1, 3, 6, and 12 months. Holter-ECG was performed at 3 months (24 h), 6 months (24 h), and 12 months (7 day) or in case of symptoms	AI—3 (6%) Ctrl—10 (20%)	N/A
Solimene 2018	3 months	AI—14 ± 6 Ctrl—12 ± 5	Clinical, 12-lead ECG and Holter	AI—17 (11%) Ctrl—16 (13%)	N/A
Casella et al. ²⁰	90 days	24	Follow-up at 3, 6, 12, 18, 24 months. 24 h Holter ECG was performed before each visit	AI—11 (15.3%) Ctrl—22 (30.6%)	Continued and then stopped at follow-up. AI—45 (62.5%) Ctrl—31 (43%) Off AADs at 24 months
Dhillon et al. ²¹	3 months	12	12-lead ECG and 48-h Holter	AI—3 (6%) Ctrl—17 (34%)	N/A
Kobori et al. ²²	3 months	N/A	N/A	N/A	N/A
Lee et al. ²³	N/A	N/A	N/A	N/A	N/A
Parollo et al. ²⁴	3 months	12	N/A	AI—5 (12%) Ctrl—7 (17%)	N/A

AAD, anti-arrhythmic drug; AF, atrial fibrillation; AI, ablation index guided catheter ablation group; AT, atrial tachycardia; Ctrl, control group (non-ablation index guided catheter ablation); ECG, electrocardiogram.

Discussion

Our results suggest that AI-guided catheter ablation is associated with lower incidence of AF recurrence, with a safety profile comparable to non-AI catheter ablation.

We also found that AI-guided catheter ablation is associated with a significant reduction in procedural, fluoroscopy, and ablation time. Furthermore, PV isolation after single encirclement was more frequent in the AI-guided radiofrequency ablation group, while acute PV reconnection was more frequently observed in non-AI-guided ablation. High heterogeneity observed for procedure duration, fluoroscopy duration and first-pass isolation can be explained by different work-flows used in different centres, differences in operator skill, speed, and need for screening. The variability in the performance of operators and centres performing radiofrequency ablation has been previously discussed.²⁵

With regards to safety, the incidence of events was low in both treatment groups, and this systematic review was not powered to show significant differences between groups. Also, not enough data

were available to compare the incidence of complications across studies depending on AI settings. However, the rate of pericardial complications was numerically lower (nearly half) in the AI-guided approach. Importantly, there were no aorto-oesophageal fistulas reported in any of the patients who underwent AI-guided ablation, and this was despite the use of high power in the backwall (power ≥ 35 W) in the two studies.

No randomized-controlled trials were identified, with evidence for the use of AI comes from observational and non-randomized studies. However, these low- moderate-quality data suggest a clear efficacy benefit and comparable safety.

The studies included used a variety of AI settings, and these settings were available in five of the six studies that demonstrated a reduction in AF recurrence compared to those who underwent non-AI guided ablation. The most commonly used energy was 35 W to the anterior and roof segments, and 30 W to the posterior and inferior segments. The most commonly used AI target was 500–550 for the anterior segment, 400–450 for the roof, posterior and inferior segments. Importantly, Philips et al.¹⁸ demonstrated that high power,

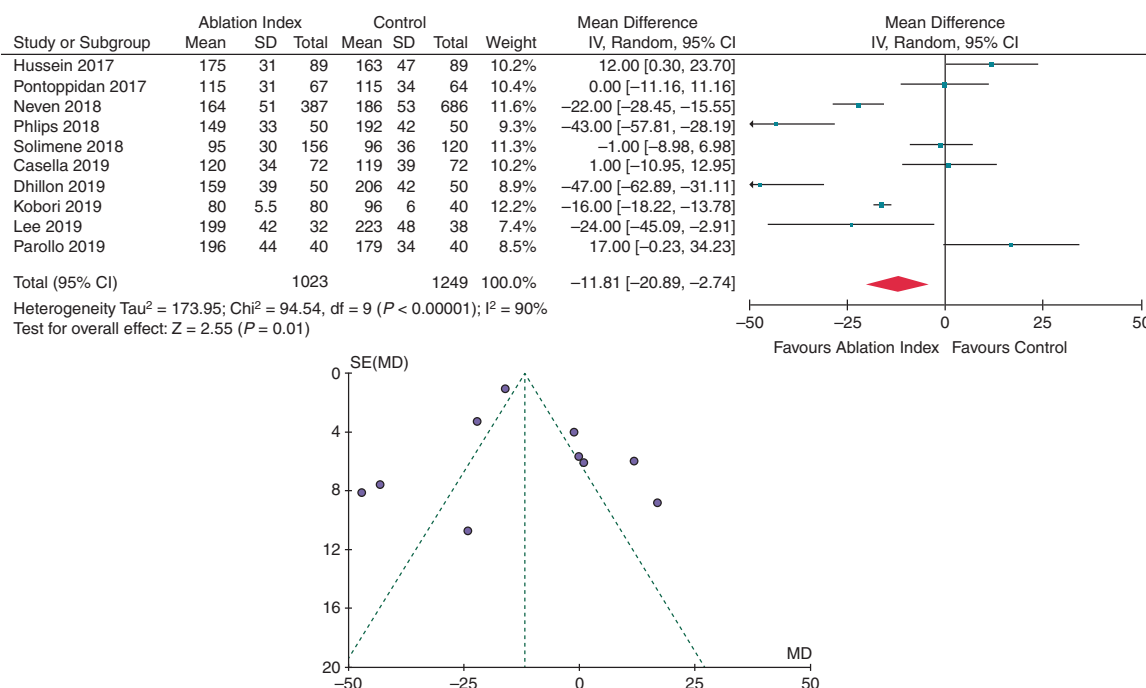


Figure 2 Forest plot comparing the total procedural time in minutes in AI-guided ablation vs. non-AI-guided ablation (control), and the corresponding funnel plot. AI, ablation index.

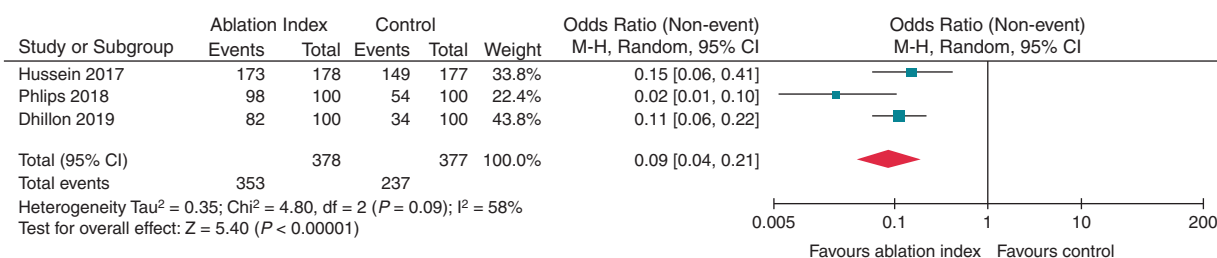


Figure 3 Forest plot comparing the rate of first-pass PVI in AI-guided ablation vs. non-AI-guided ablation (control). AI, ablation index; PVI, pulmonary vein isolation.

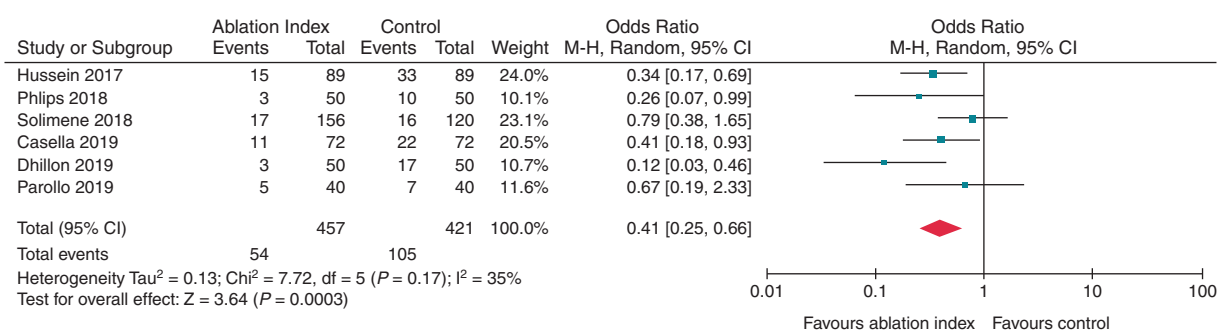


Figure 4 Forest plot comparing the effect of AI-guided ablation vs. non-AI-guided ablation (control) on freedom from AF and AT post-blanking period in patients undergoing AF ablation. AF, atrial fibrillation; AI, ablation index; AT, atrial tachycardia.

short-duration lesions using up to 50 W with AI targets of 550 on the anterior wall and 400 on the posterior wall can be effective in achieving PVI safely and efficiently.

Up until recently conventional ablation methods mainly involved the use of CF, but the availability of CF-sensing catheters has not translated to the expected improved outcomes. This is partly due to lesion creation not only being dependent on contact but also the delivery of power and duration.^{6,7} Hence focus subsequently moved to FTI, which combines force and time. However, FTI also has significant drawbacks. Force-time integral assumes a linear association between CF and time, but in reality the relationship between these factors is far more complex. Furthermore, FTI does not account for power delivery.^{8–10} Ablation index (AI) potentially overcomes the limitations of CF and FTI by incorporating power, CF and time in a weighted formula in order to predict lesion quality, and has been shown to reliably predict lesion depth in an animal model.²⁵

Hussein et al.¹⁴ showed that AI-guided ablation is associated with significant improvements in the incidence of acute PVR and atrial arrhythmia recurrence compared to non-AI guided ablation. This is potentially due to better-quality lesions as suggested by a greater drop in impedance. Ablation index can not only achieve low rates of PVR but also help to shorten the time to achieve acute PVI with a high rate of first-pass isolation. Data suggest that an optimal AI-targeted PVI is generally feasible and could improve acute outcomes in patients with AF.^{14,18,21}

Dhillon et al.²¹ demonstrated that AI is not only effective but also reproducible, when a pre-specified protocol is followed respecting strict criteria for contiguity and quality lesion, resulting in high and comparable rate of acute PV isolation among different operators using different catheters, AI settings, and fluoroscopy times.²⁶

Although the safety of AI has not been widely evaluated, PV ablation guided by AI appears to have a low complication rate and leads to a higher single-procedure arrhythmia-free survival at 12 months.^{15,16,18}

A recent meta-analysis of PVI prior to the use of the AI in 1774 patients with paroxysmal AF demonstrated success rates of 78% after 12 months, which is slightly lower than we observed for paroxysmal AF patients (>90% freedom from atrial arrhythmia relapse at 12 months).²⁷ Success rates are lower in patients with persistent AF, with a meta-analysis of 18 657 patients reporting a success rate of only 43% following a single procedure.²⁸ Our study had a mixed population of paroxysmal and persistent AF and no study included only persistent AF patients, and hence we are not able to provide a precise figure on this. However, Hussein et al.¹⁴ have reported this to be close to 80% at 12 months.

Limitations

Our meta-analysis has limitations that should be acknowledged including the following: (i) the available studies are not randomized-controlled trials and as such are not high quality, (ii) high heterogeneity among included studies for some of the endpoints like procedural and fluoroscopy duration. However, this probably reflects differences in the ablation approach and usage of screening by the different operators as explained above, (iii) small sample size of studies, (iv) different AI values for different sites of the atrium, (v) follow-up was not consistent across studies, with 7-day Holter not always used; furthermore, the ILR was not used in any of the studies, and hence these

may have underestimated the relapse rate; also, comparison of arrhythmia burden with both strategies would be of interest, (vi) data concerning the ablation settings used in control groups, such as the interlesion distance and catheter stability, were not always available, therefore, we cannot exclude that the observed improvements potentially related to improvements in all these factor and not just AI.

Conclusions

Low to moderate quality of evidence from non-randomized data suggests that AI-guided catheter ablation is associated with a significantly lower incidence of AF recurrence, while it has comparable safety profile to non-AI catheter ablation. Also, the AI-guided approach was significantly associated with lower procedure times (total, screening, and ablation), more frequent first-pass isolation and lower rate of acute PV reconnection. A large randomized-controlled trial is required confirm the benefits of AI-guided catheter ablation.

Supplementary material

Supplementary material is available at *Europace* online.

Conflict of interest: none declared.

Data availability

The data underlying this article are available in the article and in its online [Supplementary material](#).

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